

Hit-Competition Study at High Track Density

Derived from `hit_competition_study.ipynb`

Quantum Track Reconstruction

April 2, 2026

Abstract

This document summarizes the notebook `Toy_Characterisation/hit_competition_study.ipynb`, which tests whether the Hamiltonian coupling term γ suppresses false competing segments as track density increases. The study proceeds through six stages: (1) hit occupancy, (2) activation spectrum separation, (3) per-hit competition behavior, (4) track-level reconstruction quality, (5) threshold-sensitivity curves, and (6) segment-level scattering sweeps. The workflow is designed for high-density stress testing and saves all artifacts by run timestamp under `hit_competition_outputs/`.

1 Goal and Hypothesis

The classical solve of the linear system $Ax = b$ yields segment activations $x_i \in [0, 1]$. Segments sharing a hit are coupled through the Hamiltonian. The central hypothesis is:

As occupancy increases, true segments remain near or above the operating threshold, while false competitors are suppressed by the competition structure encoded in A .

The notebook evaluates this mechanism at multiple event densities and angular phase-space settings.

2 Configuration and Definitions

2.1 Detector and Event Generation

The notebook uses a 5-module planar geometry with module spacing $\Delta z = 33$ mm and symmetric angular generation windows $\phi_{\max} = \theta_{\max} \in \{0.2, 0.1, 0.04\}$ rad. A density sweep is performed over

$$N_{\text{tracks}} \in \{5, 10, 20, 30, 50, 75, 100, 150\}$$

with repeated event generation per point.

2.2 Hamiltonian Parameters

The baseline parameter set is

$$\sigma_{\text{res}} = 0, \quad \sigma_{\text{scatt}} = 10^{-4} \text{ rad}, \quad \gamma = 1.5, \quad \delta = 1.0,$$

and

$$\varepsilon = \sqrt{2\theta_s^2 + 12\theta_r^2 + 2\theta_{\min}^2}, \quad \theta_s = s \sigma_{\text{scatt}}, \quad \theta_r = \arctan\left(\frac{s \sigma_{\text{res}}}{\Delta z}\right),$$

with scale $s = 3$. The activation operating point is

$$\text{baseline} = \frac{\delta}{\delta + \gamma}, \quad \text{threshold} = \frac{1 + \text{baseline}}{2}.$$

3 Analysis Workflow

3.1 Step 1: Hit Occupancy

For each generated event, the study builds a map `hit_id` $\rightarrow$ candidate segment indices and records occupancy statistics (mean, median, max, spread). This quantifies combinatorial pressure and validates the expected scaling for internal layers.

3.2 Step 2: Activation Spectrum (True vs False)

Each candidate segment is labeled true/false from ground-truth track adjacency. The solved activations x_i are then compared between classes, including:

- fraction of true segments above threshold,
- fraction of false segments above threshold,
- separation statistic from pooled variance.

This step directly measures whether ambiguity suppression degrades at high occupancy.

3.3 Step 3: Per-Hit Competition Curves

For contested hits (shared by multiple segments), the notebook compares:

- true segment activation x_{true} ,
- summed false competitor load $\sum_j x_{\text{false},j}$,
- number of competitors.

Scatter and summary curves test whether true segments remain robust under increasing local competition.

3.4 Step 4: Track-Level Reconstruction Quality

An end-to-end chain `generate` $\rightarrow$ `construct Hamiltonian` $\rightarrow$ `solve` $\rightarrow$ `get_tracks` $\rightarrow$ `validate` is run for each density and angle. Metrics include:

- reconstruction efficiency,
- ghost rate,
- clone fraction.

Truth and reconstructed events are also exported as JSON snapshots.

3.5 Step 5: Threshold Sensitivity (ROC-like)

Thresholds are swept over

$$\tau \in [0.2, 0.9]$$

at representative densities. For each τ , efficiency and ghost rate are computed, giving operating curves and highlighting the default threshold point.

3.6 Step 6: Segment-Level Scattering Sweep

A second scan varies multiple-scattering strength by factors $1\times$, $2\times$, and $4\times$ (with corresponding ε values), across event sizes 10 to 100 tracks. Reported metrics:

- segment efficiency,
- segment false rate,
- true/false segment-pair counts,
- accepted true/false pairs with $\theta \leq \varepsilon$.

Additional histograms compare true and false pairwise-angle distributions for fixed-size high-density ensembles.

4 Produced Artifacts

Each run creates a timestamped directory:

```
hit_competition_outputs/<RUN_TAG>/{data,figures,events}
```

Typical figures (saved when corresponding steps are executed):

- step1_occupancy_comparison.png
- step1_occupancy_histograms.png
- step2_activation_spectra.png
- step2_activation_summary.png
- step3_competition_scatter.png
- step3_competition_summary.png
- step4_reco_quality.png
- step5_roc_by_angle.png
- step5_threshold_curves.png
- step6_scattering_comparison.png
- step6_acceptance_histograms.png

5 Representative Figures

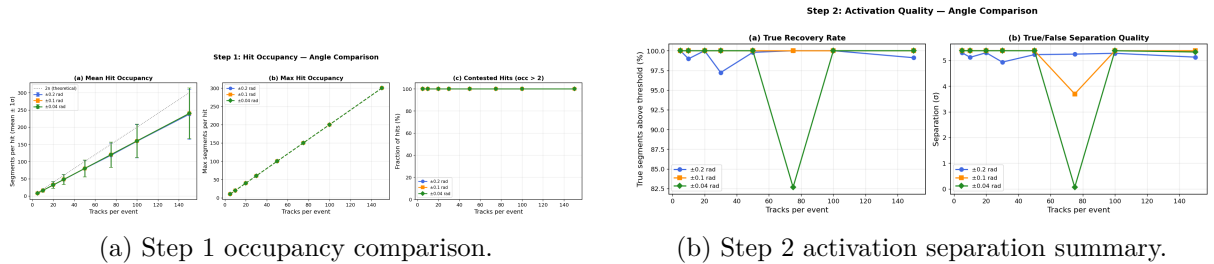
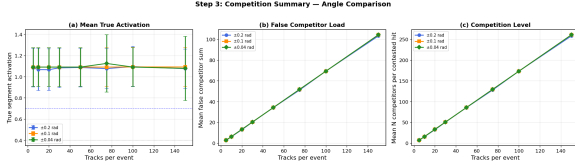
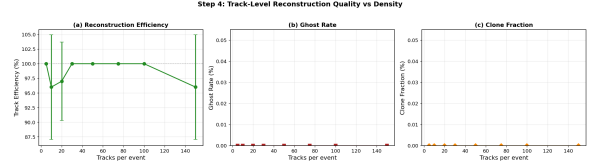


Figure 1: Occupancy and activation-separation diagnostics.

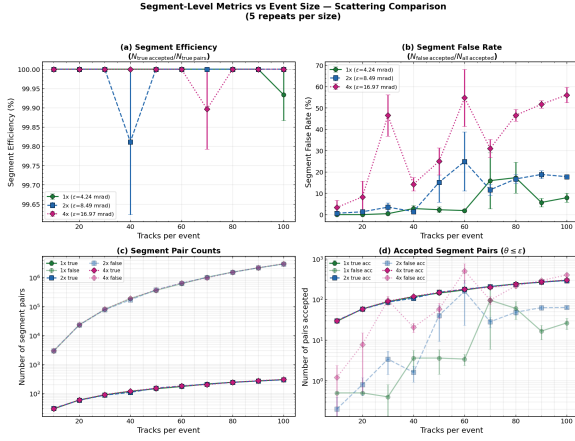


(a) Step 3 per-hit competition summary.

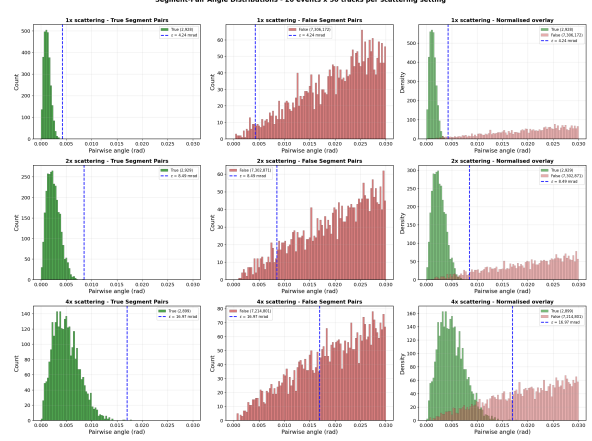


(b) Step 4 track-level quality.

Figure 2: Competition behavior and end-to-end reconstruction quality.



(a) Segment efficiency reference plot.



(b) Segment-angle histogram reference plot.

Figure 3: Additional segment-level reference outputs available in the study directory.

6 Interpretation Guide

The notebook supports the following checks:

1. **Combinatorics scaling:** occupancy trends with density and angle indicate where overlap becomes severe.
2. **Activation robustness:** true/false spectrum separation indicates whether competition suppression remains effective.
3. **Operational envelope:** ROC-like curves identify threshold headroom as density increases.
4. **Physics stress response:** scattering sweeps quantify sensitivity of segment-level acceptance and contamination.

7 Reproducibility Notes

- The notebook can either compute fresh results or load a previous run via `LOAD_RUN`.
- All JSON summaries are saved in the run's `data/` directory.
- To display figures from another run, update `\runfigdirA` and `\runfigdirB` at the top of this file.

8 Conclusion

This study provides a structured, multi-level validation of hit-competition handling in the Hamiltonian reconstruction approach. By combining local (occupancy and per-hit competition), intermediate (segment activations), and global (track-level metrics and threshold scans) diagnostics across density, angle phase space, and scattering strength, the notebook establishes a clear framework for quantifying high-density robustness.